



PMI-ACP® EXAM PREP

PMI Authorized
Training Partner

BOOTCAMP Session 6

- This course will assist learners in preparing for PMI's PMI-ACP Exam (2024 Update)

ATTENDANCE TRACKING

Percipio Users:

Name is based on your log in information in Percipio

Using Zoom:

Enter your first and last name

BREAKS



Yes! We will have periodic breaks

For attendance purposes, please stay logged in during all breaks.



CHAT vs Q&A

Please use the **Chat** for:

- **Greetings** before the session starts and during breaks
- Once the session starts , the chat may be closed or changed to *Hosts & Panelists Only* to minimize disruptions and focus on important information.
- The instructor may open the chat during the session for student **to respond to the instructor's questions** and create a group dialog.

CHAT vs Q&A

Please use the **Q&A** for:

- **Technical assistance** – Begin with: Percipio or Non-Percipio student
- Guidance on how to **access course material** – Begin with: Percipio or Non-Percipio
- Clarification and **questions on lecture points**, if not answered by instructor
- The Q&A may be open and closed throughout the session to allow us to address questions/issues in a timely manner.
- **Please be very patient, the support team responds to many inquiries per session**

IS LIVE ATTENDANCE REQUIRED?

- **YES**, if you are taking this training to register for the PMI-ACP exam
- You are **allowed to miss up to two sessions IF** you make up the sessions by **watching the video replays**.
- A **missed** session means you were disconnected for **more than a total of 15 mins** for the duration of the session.
- If you **miss three or more sessions**, you will need to make up the missed time by **attending live in another 8-day cohort**.
- *Please see the Bootcamp Calendar for information about upcoming sessions at: <http://calendar.skillsoft.com/>



ACCESSING THE VIDEO REPLAYS

1. Go to: <https://github.com/Skillsoft-Content/PMI-ACP-Replay>
2. Replays will be available within 2 business days after the session ends.
3. Click on the Excel file for the year you attended the Bootcamp. You won't see a *file open* option, but it is selected.
4. Click the *Download raw file* button on the far left-hand side.
5. Open the downloaded file using this password: **acpB00tcampReplay!**



Those are zero's not the letter O. The password is case sensitive.

6. Locate and open the worksheet tab that corresponds with the bootcamp you attended
7. Make a note of the passcode.
8. Paste the provided link into your browser.
9. Complete the required registration steps
10. Input the passcode when prompted
The password to open the Excel file is NOT the passcode to access the replay.

Note: Replays will be available for 1 year.
They are not available for download.

No limit to watch replays to study



PMI®-Authorized PMI Agile
Certified Practitioner (PMI-ACP)®
Exam Preparation Course

Lesson Three

Product

Version 1.0 | 2024 Release

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Manage increments



Refine the
product backlog



**Manage
increments**



Visualize work



Manage value
delivery





Measure the delivery of value

Section 4 of 4

Measuring results

Empirical and value-based measurements

Demonstrable value

Learning while delivering value



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Measures of value

CSAT

NPS

Smiley chart

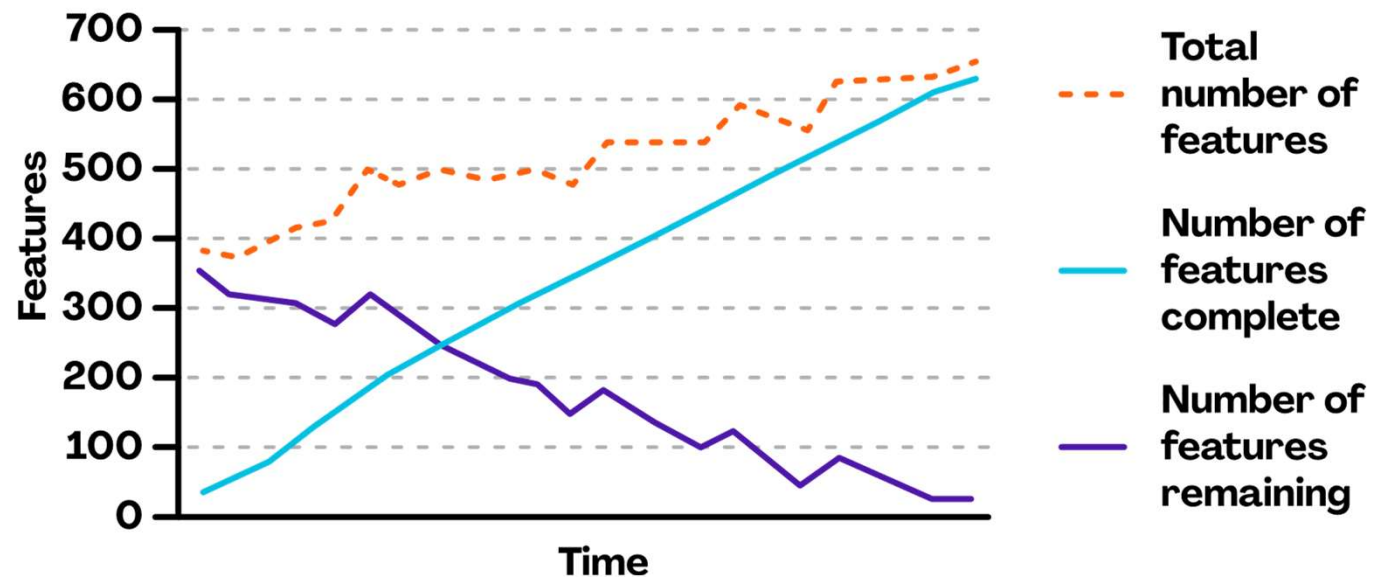
Burndown chart

Burnup chart

Cumulative flow diagram

Feature chart

Features complete, remaining, and total



Assessing value



Return on
investment (ROI)



Internal rate of
return (IRR)



Net present
value (NPV)

ROI

Ratio of the benefits received from an investment

Money invested in it

X 100

Expressed as a percentage

Example: $120/12 = 10$

NPV

$$\frac{\text{Present value of a revenue stream (income - costs)}}{\text{Number of time periods}}$$

Example: Present value of \$1,500 over five years = 300

IRR

What it is:

IRR is the specific interest rate that makes the present value of an investment's future cash inflows equal to its initial cost (cash outflow).

What it shows:

A 10% IRR means the investment is expected to yield a 10% annual return over its life.

How it's used:

It helps investors and businesses evaluate potential projects. An investment is considered worthwhile if its IRR is greater than the company's required rate of return.

Using EVM with adaptive projects

Pros	Cons
EVM is a leading indicator	Plans change, which removes the basis for effective EVM
Visual indicator	EVM isn't a true indicator of project success

Visualize work



Refine the
product backlog



Manage
increments



Visualize work



Manage value
delivery



Educate team on work visualization techniques

Kanban boards

Pull system

WIP limits



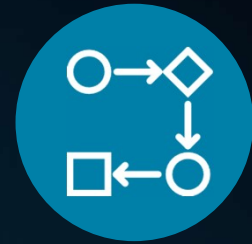
General practices of Kanban



**Visualize the
workflow**



**Limit WIP
(work in progress)**



Manage flow



**Make policies
explicit**



**Implement
feedback loops**



**Improve collaboratively,
evolve experimentally**



Source:
Revisiting the Principles and General Practices of the Kanban Method - David J. Anderson School of
Management. <https://djaa.com/revisiting-the-principles-and-general-practices-of-the-kanban-method/>

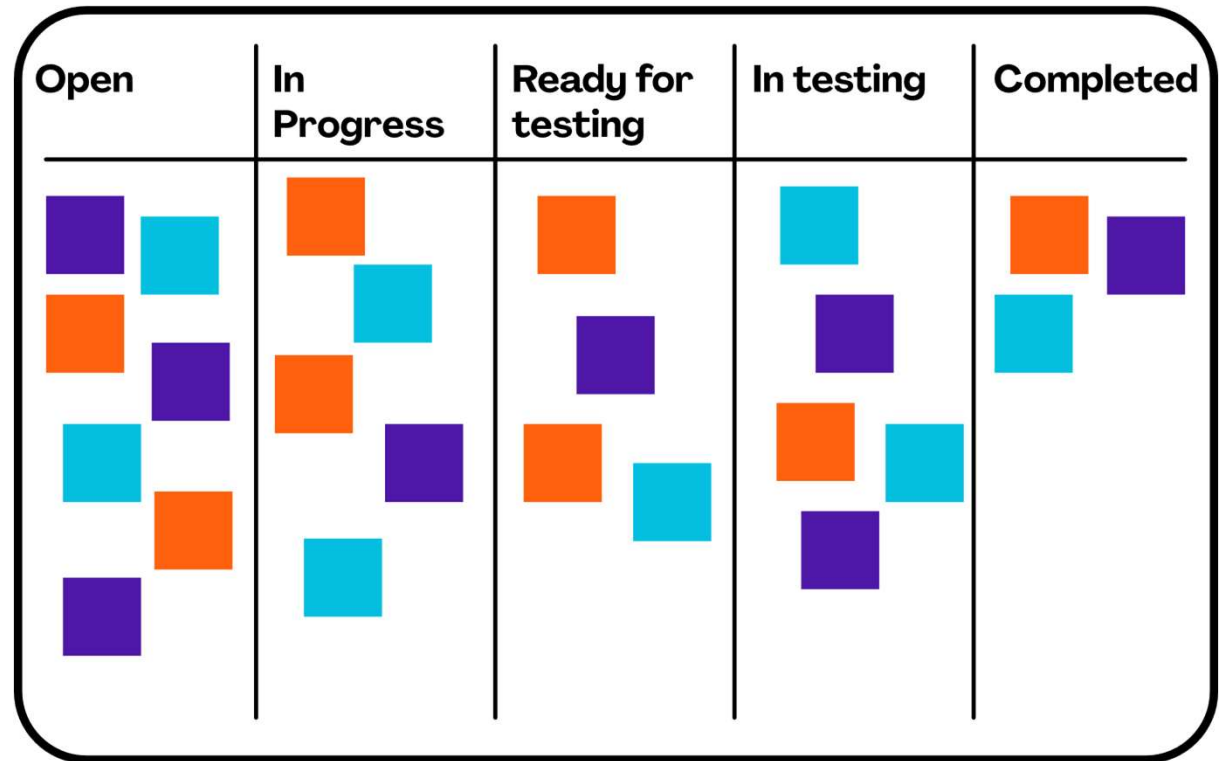
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Task boards

Convenient mechanism

Necessary flexibility

Large board



Source:
Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp 221–227.

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Task board makeup

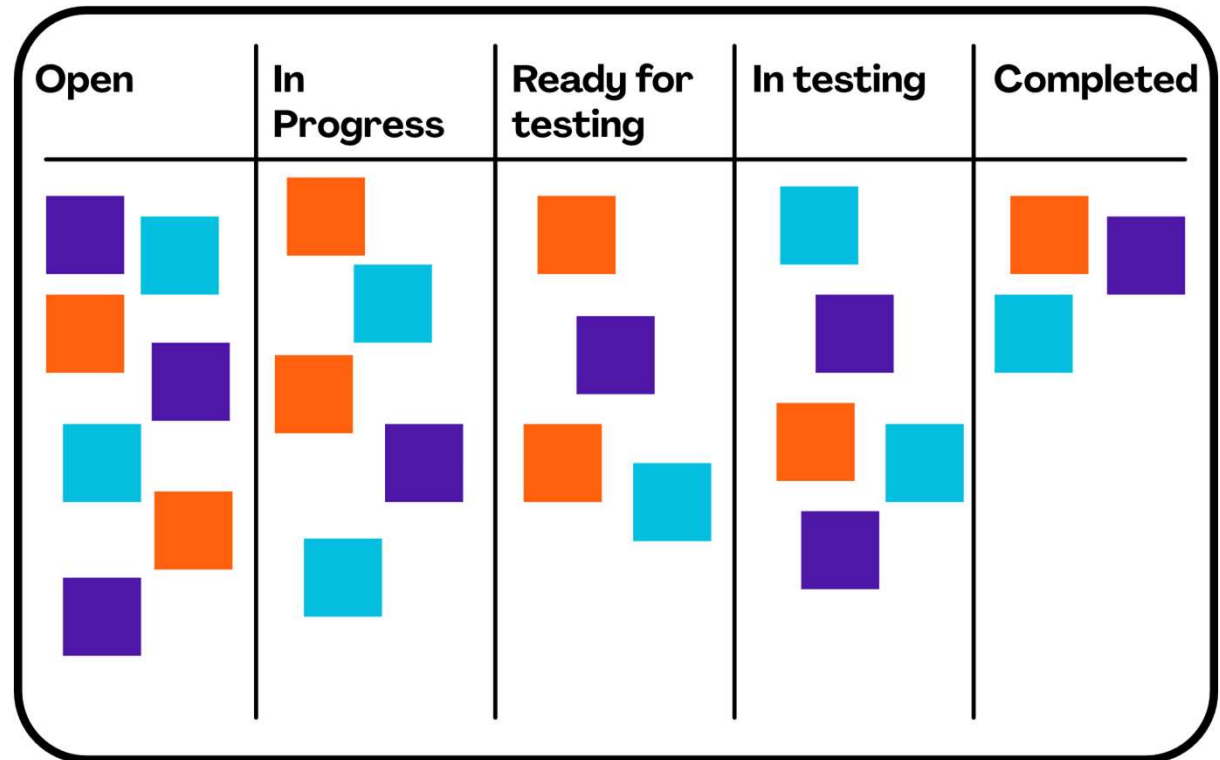
Open

In progress

Ready for testing

In testing

Completed



Source:
Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp 221–227.

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Work item creation

Information to include:

Necessary to do the work

Interesting to others

Different types of work

Blockers?

Source:

Hammarberg, M. & Joakim Sundén. (2014). *Kanban in Action*. Manning. pp. 83–98.

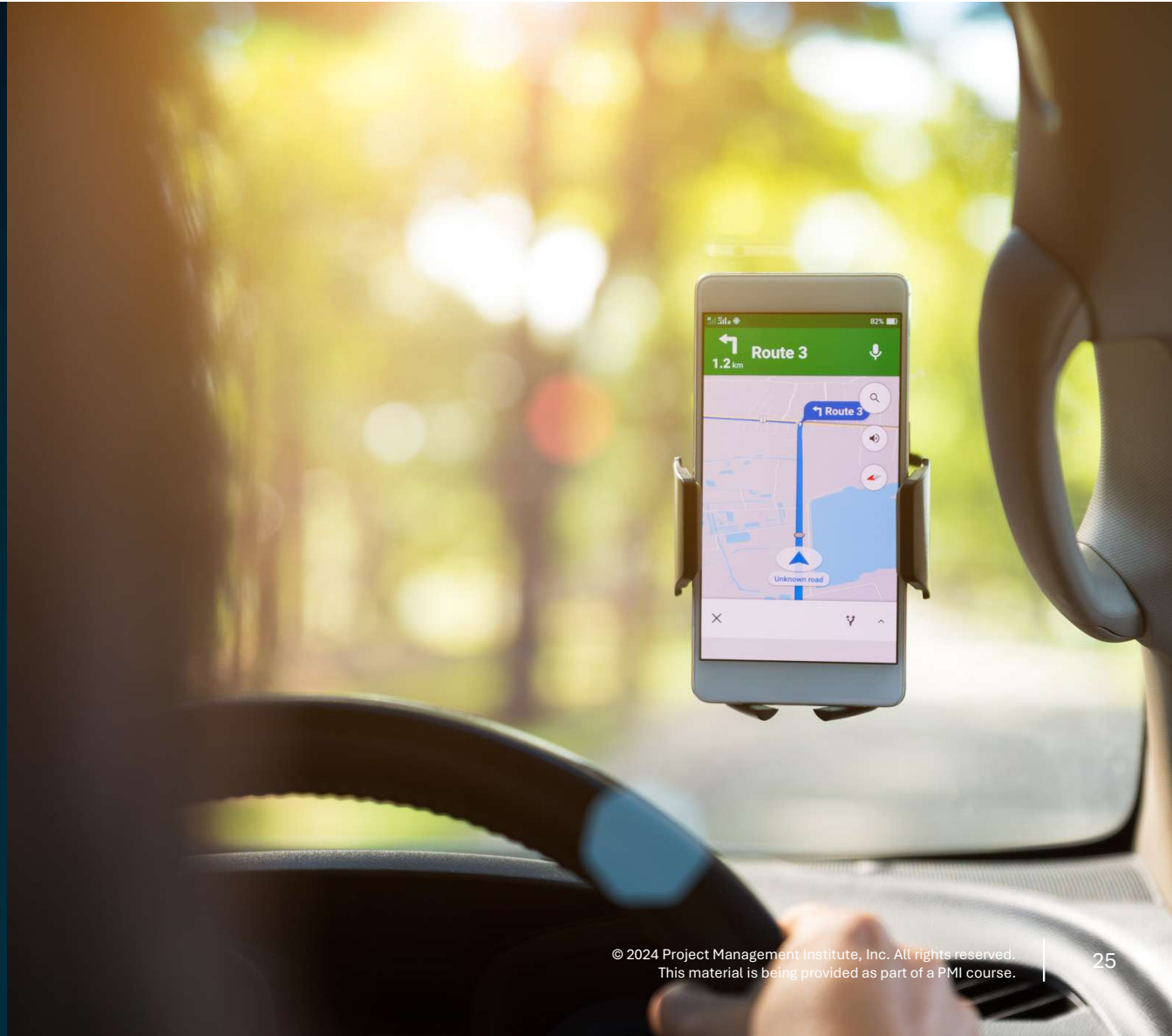
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Tracking effort

Evaluation apprehension

Not always easy

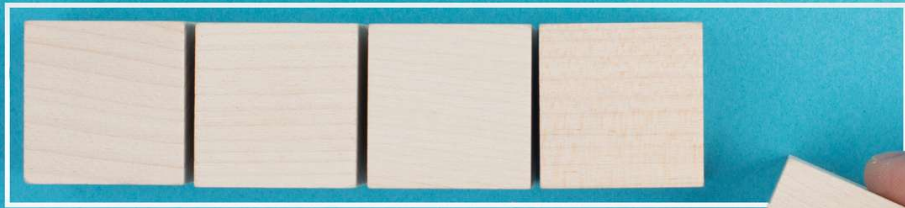
Variability



Source:
Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp 221–227

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Progress...



Progress indicator

Tool that helps track “how much done” the item is at a point in time.

Source:

Hammarberg, M. & Joakim Sundén. (2014). *Kanban in Action*. Manning. pp. 83–98.

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Velocity

Number of units completed

Velocity cautions



Source:
Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp 221–227.



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Personas

Descriptions of
fictional people who fill
roles as stakeholders

PM tools

Visual tools

Software tools

Which ones does your
company use?



Source:
Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp 221–227.



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Manage value delivery



Refine the
product backlog



Manage
increments



Visualize work



**Manage
value delivery**





Define what success will look like (e.g., success criteria, sustainability, security, privacy, regulatory, compliance)

Section 1 of 2

Defining success factors

Meet goals

Conditions of satisfaction

Date-driven

Feature-driven



Source:

Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp. 133



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Definitions of success

Definition of ready

Full Kit

Definition of done (DoD)

Definition of broken

Source:

Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp 133.

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Successful criteria for agile approaches

Shipped

Team intact

Team would work same
way again



Source:
Cockburn, A. (2009). *Agile Software Development: The Cooperative Game*.
Addison-Wesley.



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Success patterns

Face-to-face
communication

Excess weight is costly

Larger teams =
heavier approaches

Greater criticality =
greater ceremony

Intermediate deliverables

Process is not discipline

Efficiency is expendable

Source:
Cockburn, A. (2009). *Agile Software Development: The Cooperative Game*.
Addison-Wesley.

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Ensure the value increments are optimized

Section 2 of 2



Prioritization

Responsibility shared
by team

Stories and themes

Tangible value



Source:

Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp. 79–88.

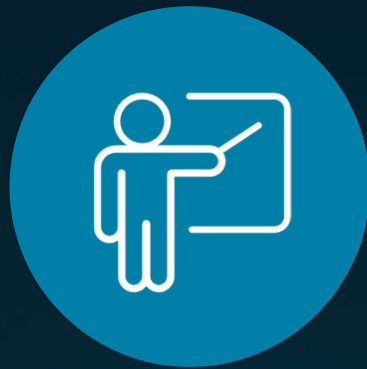


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Prioritization factors



Value



Cost



New knowledge



Risk



Source:

Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp. 79–88.

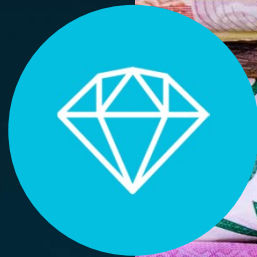
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Financial value

How much money will the organization make or save?

Impact over a period

Difficult to estimate



Source:

Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp. 79–88.

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Cost

Major factor in priority

Often viewed in terms of time



Cost of a feature
over a time period =
Salaries of all team members
Story points completed

Source:

Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp. 79–88.

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New knowledge

Knowledge about the customer

Knowledge about the product

Knowledge about the project

Reduce uncertainty



Source:

Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp. 79–88.

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Risk

Schedule risk

Budget risk

Scope risk

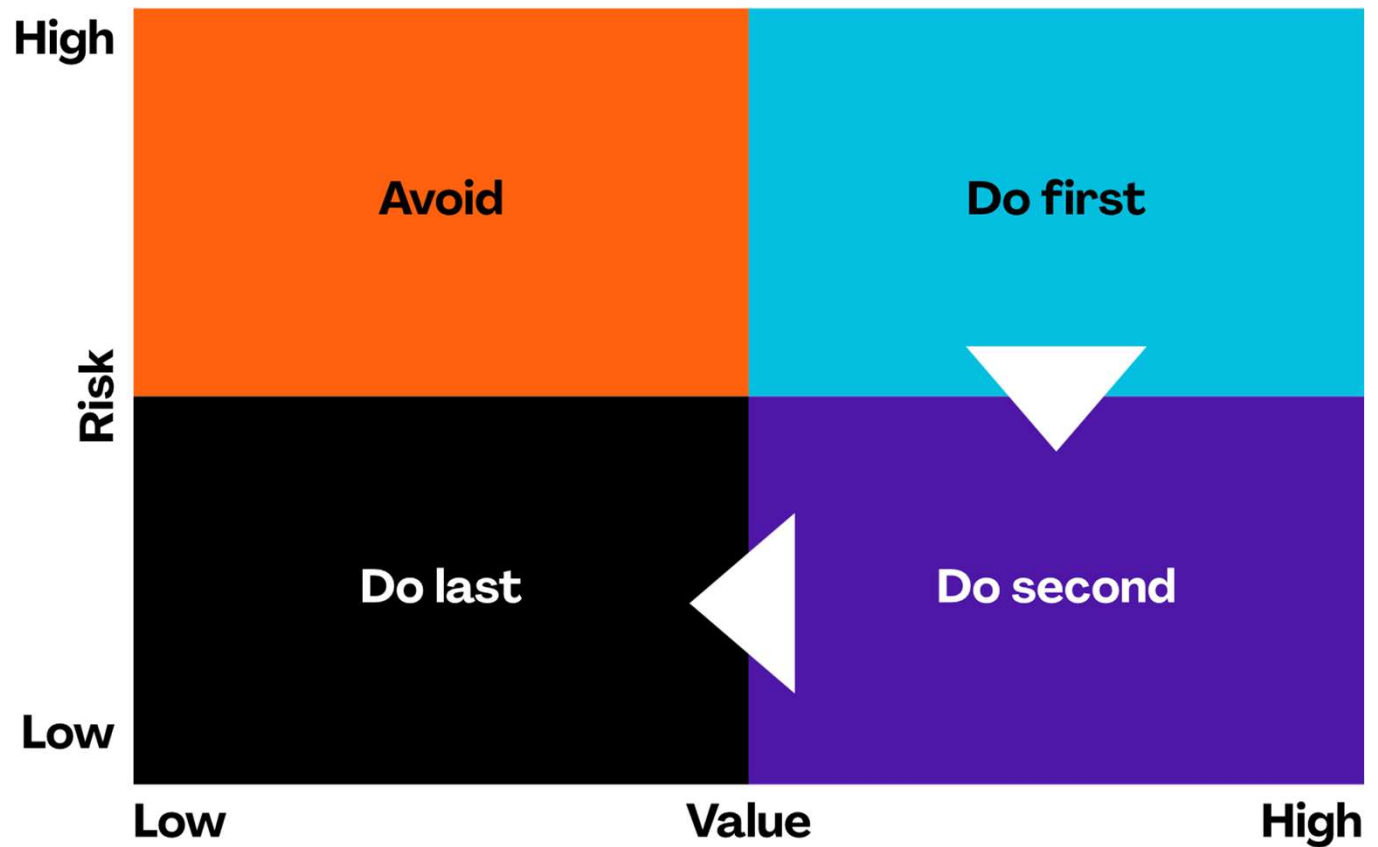


Source:

Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp. 79–88.

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Relationship between risk and value



Source:
Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp. 79–88.

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Combining these factors

Consider feature value

Move forward or backward

Source:

Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp. 79–88.

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Infrastructure example

Value

Cost

Knowledge

Risk



Source:

Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp. 79–88.

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User interface design example

Value

Cost

Knowledge

Risk



Source:

Cohn, M. (2005). *Agile Estimating and Planning*. Prentice Hall. pp. 79–88.

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Lesson Four

Delivery

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During this segment



**Seek early
feedback**



**Manage
agile metrics**



**Manage
impediments
and risk**



**Recognize
and eliminate
waste**



**Perform
continuous
improvements**



**Engage
customers**



**Optimize
flow**



**Evaluate
customer
satisfaction**



**Deliver work in
small increments**



**Collect and
incorporate
stakeholders'
feedback on a regular
basis**



**Seek early
feedback**



**Manage
agile metrics**



**Manage
impediments
and risk**



**Recognize
and eliminate
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**Perform
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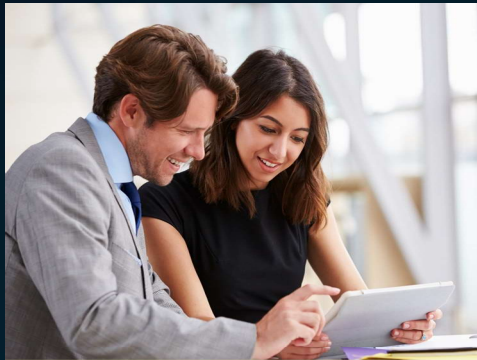


Evaluate customer satisfaction

Section 1 of 3

Principles of the Agile Manifesto (1–4)

How do you evaluate whether the customer is satisfied?



- 1 Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.



- 2 Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.



- 3 Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.



- 4 Business people and developers must work together daily throughout the project.



Source: Beck, K., Beedle, M., van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., Grenning, J., Highsmith, J., Hunt, A., Jeffries, R., Kern, J., Marick, B., Martin, R. C., Mellor, S., Schwaber, K., Sutherland, J., & Thomas, D. (2001). *Principles behind the agile manifesto*. Manifesto for Agile Software Development. <https://agilemanifesto.org/principles.html>

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Make sure the product owner and stakeholders attend

This is a prime opportunity to regularly gauge their satisfaction

Consider inviting customers to demos

You'd be hard-pressed to find a better way to find out if what you're building meets their needs and expectations





What you do with their feedback will be critical to the success of your project

One of the key advantages to agile is the ability to quickly pivot as more becomes known

Giving the customer what they need—as opposed to what they may have intended—can be critical to their satisfaction

Customer satisfaction metrics



**Customer
satisfaction
score**



**Net
promoter
score®**

Customer satisfaction score (CSAT)

**“How would you rate your overall satisfaction with the
(goods/services) you received?”**



**1. Very
unsatisfied**



2. Unsatisfied



3. Neutral



4. Satisfied



**5. Very
Satisfied**

Calculating your customer satisfaction score (CSAT)

$$\frac{\text{Number of satisfied customers}}{\text{Number of survey responses}} \times 100$$
$$\frac{7}{10} \times 100 = 70$$



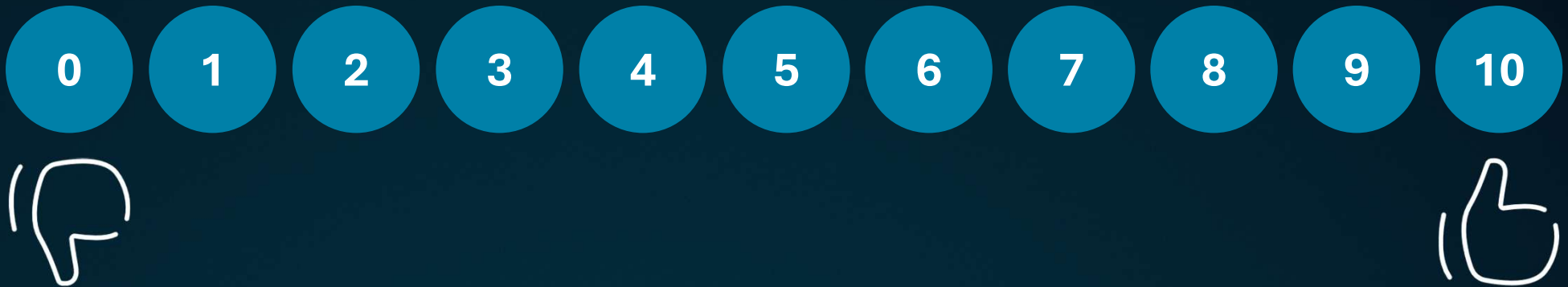
4. Satisfied



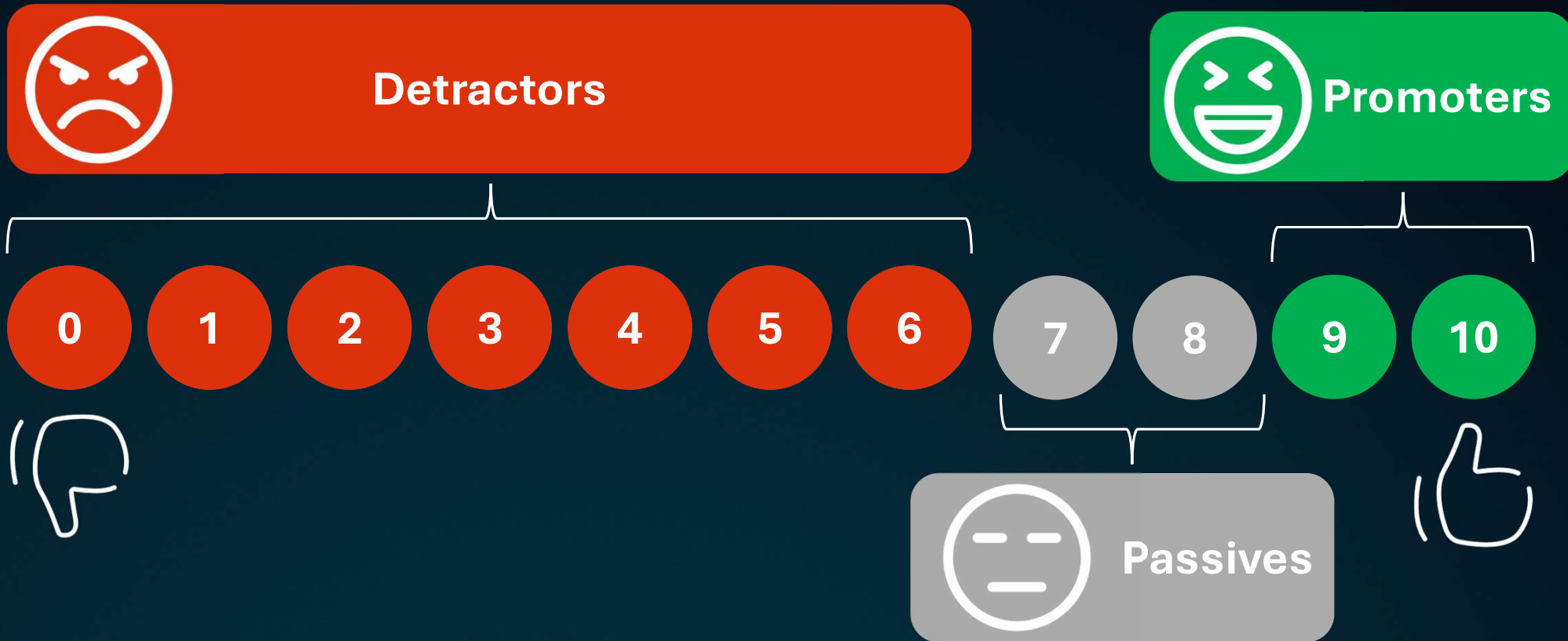
5. Very Satisfied

Net promoter score

“How likely is it that you would recommend
(organization/product/service) to a friend or colleague?”



Net promoter score categories



Calculating your net promoter score®

$$\begin{array}{ccccc} \text{😊} \% & - & \text{😡} \% & = & \text{NPS}^{\text{®}} \\ \text{Promoters} & & \text{Detractors} & & \\ 60\% & - & 20\% & = & 40 \quad \text{Net Promoter Score}^{\text{®}} \end{array}$$

Knowledge check

Which is the most frequent gauge of customer satisfaction?

A. Questionnaire

B. Pop-up

C. Iteration demo

D. Form on a website

The correct answer is the **iteration demo**. It gives you an opportunity to gauge the customer's satisfaction at the end of every iteration—and to adjust, as necessary.



Knowledge check

Seventy percent of the respondents to an NPS survey came back as promoters, 20% came back as passives, and 10% as detractors. What is your net promoter score?

A. 10

B. 60

C. 50

D. -10

The correct answer is the **60**. Subtracting the detractors (10) from the promoters (70) yields 60.





Deliver work in small increments

Section 2 of 3

Limit rework and scrap by getting feedback early

Able to get feedback early and often

That enables the team to adjust early—when it's less expensive and disruptive to do so





Constant minor adjustments

We do a little,
check to make sure we
are heading in the right
direction, then do a
little more

That's one of the most
effective ways to get to
where we want to be

We don't get too far off track

We don't get to spend so long heading in the wrong direction that we end up completely off course





Collect and incorporate stakeholders' feedback on a regular basis

Section 3 of 3

Agile approaches have short cadences

Demos are quickly followed
by retrospectives and
planning the next iteration

That's where we decide
how to act on the feedback



How do we do that?



Using scheduled events

These scheduled events allow agile teams to collect and incorporate stakeholders' feedback on a regular basis

Don't forget peripheral stakeholders



Product owner



Sponsors



Legal team



Customers



Technical personnel

That was not a complete list

The point is to
regularly collect and
incorporate feedback
from your stakeholders





**Feedback is only
helpful if we act on it
appropriately**



**Determine which
metrics are
appropriate for a
given audience**



**Radiate metrics
across the relevant
audience**



**Review and
analyze metrics**



**Use metrics
insights for
decision-making**



**Seek early
feedback**



**Manage
agile metrics**



**Manage
impediments
and risk**



**Recognize
and eliminate
waste**



**Perform
continuous
improvements**



**Engage
customers**



**Optimize
flow**



Determine which metrics are appropriate for a given audience

Section 1 of 4

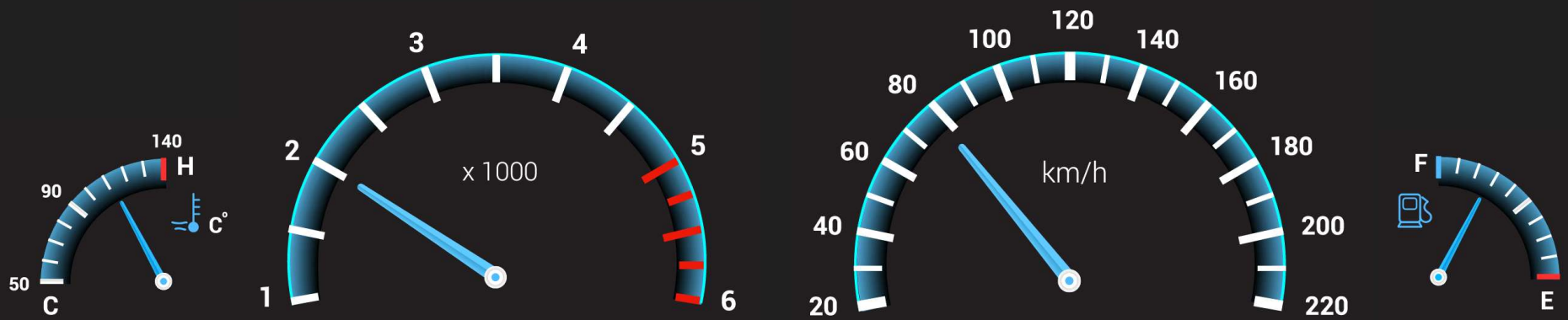




Why do we measure?

For teams to be successful, they need a clear vision, direction, and way of tracking progress and taking corrective actions as necessary

Automobile gauges provide metrics



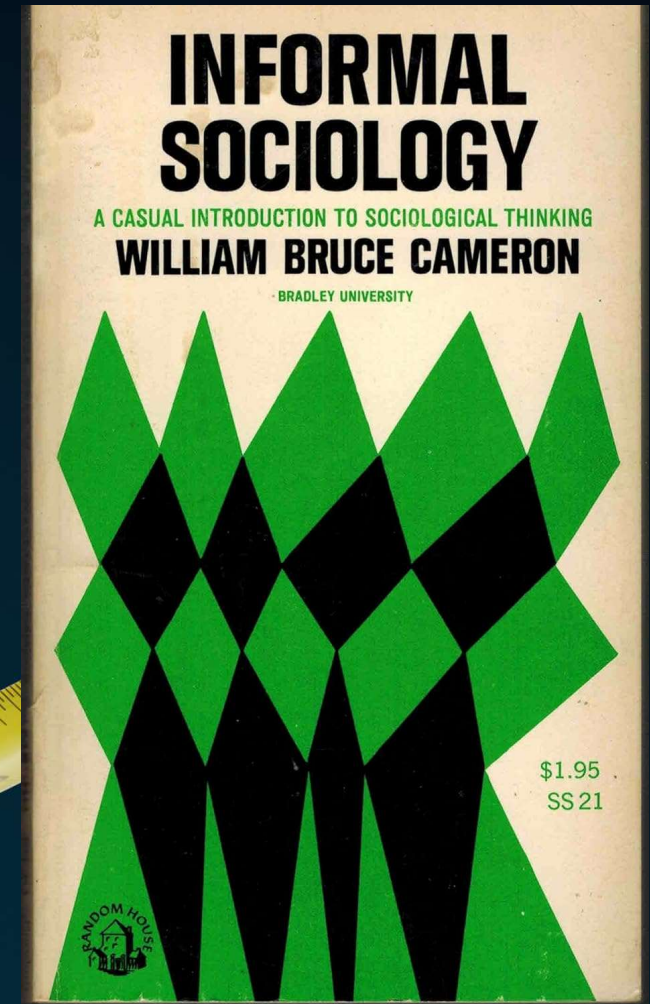
Leading versus lagging indicators

We want to focus most of our attention on what lies ahead, so we can navigate and take appropriate action



What should we measure?

“Not everything that can be counted counts,
and not everything that counts can be counted.”



Source:
Cameron, W. B. (1969). *Informal sociology: a casual introduction to sociological thinking*. Random house.

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What metrics will produce meaningful results?



Simple



**Relevant
to the end goal**



**Leading
future focused**



Source:
Reinertsen, D. (1998). *Managing the design factory:
A product developers toolkit*. Free Press.

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Checking traditional project metrics:

Budget consumed



Simple



Relevant to end goal



Leading indicator



**We want completed projects,
delighted customers, and value
delivered—not spent funds**

Checking traditional project metrics:

Conformance to plan



Simple



Relevant to end goal



Leading indicator



Tracking our progress against a faulty map will not help us reach our destination

Checking traditional project metrics:

Hours worked



Simple



Relevant to end goal



Leading indicator

Tracking hours worked can have unexpected side effects, including long hours, burnout, defects, and consumed budgets

Checking traditional project metrics:

Lines of code written



Simple



Relevant to end goal



Leading indicator



Tracking lines of code written

does not reward simplification

and can lead to code bloat



Think about it

Does your current team use any traditional metrics?

In what ways are they useful—or unuseful—to you?

What are some typical agile metrics?



**Burndown
charts**



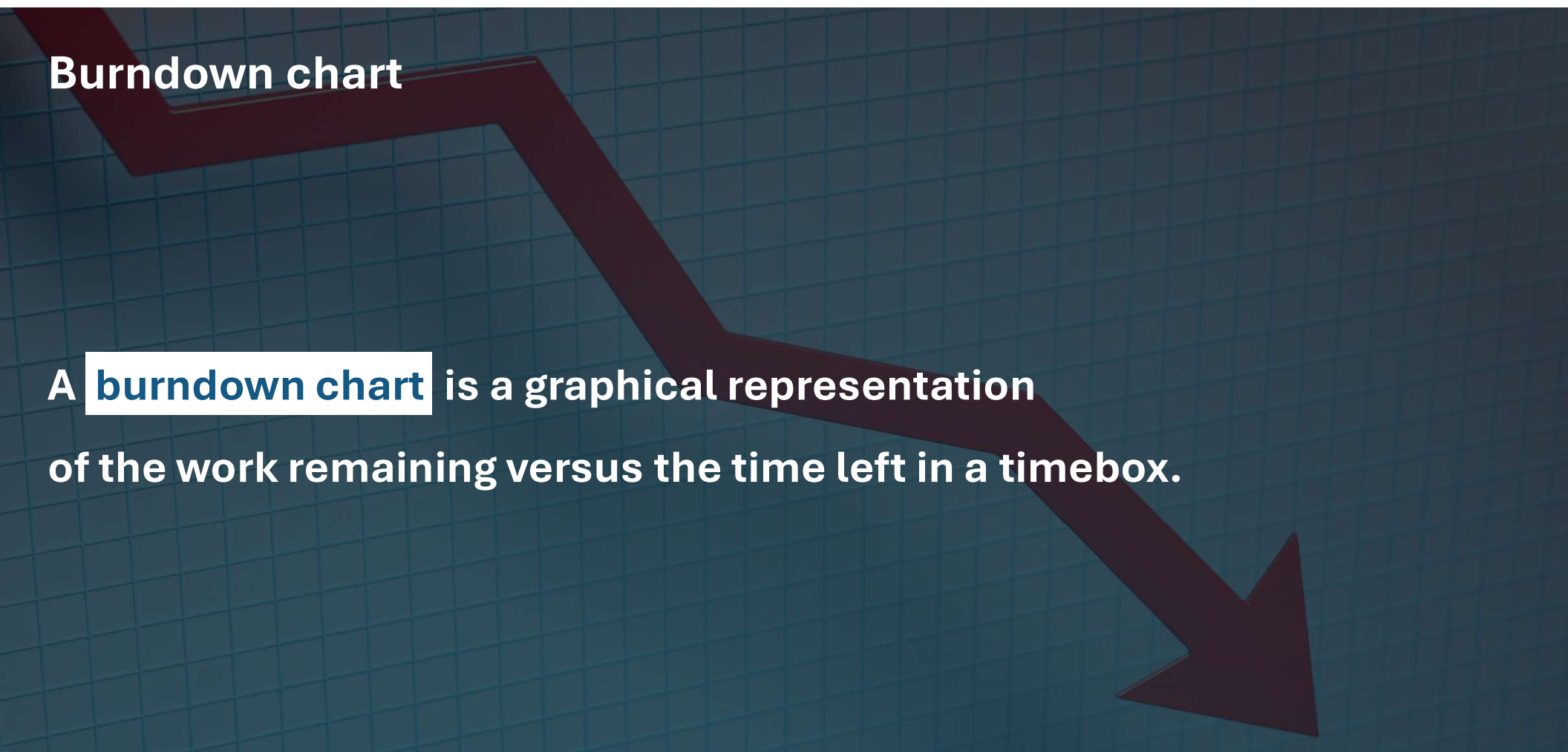
Burnup charts



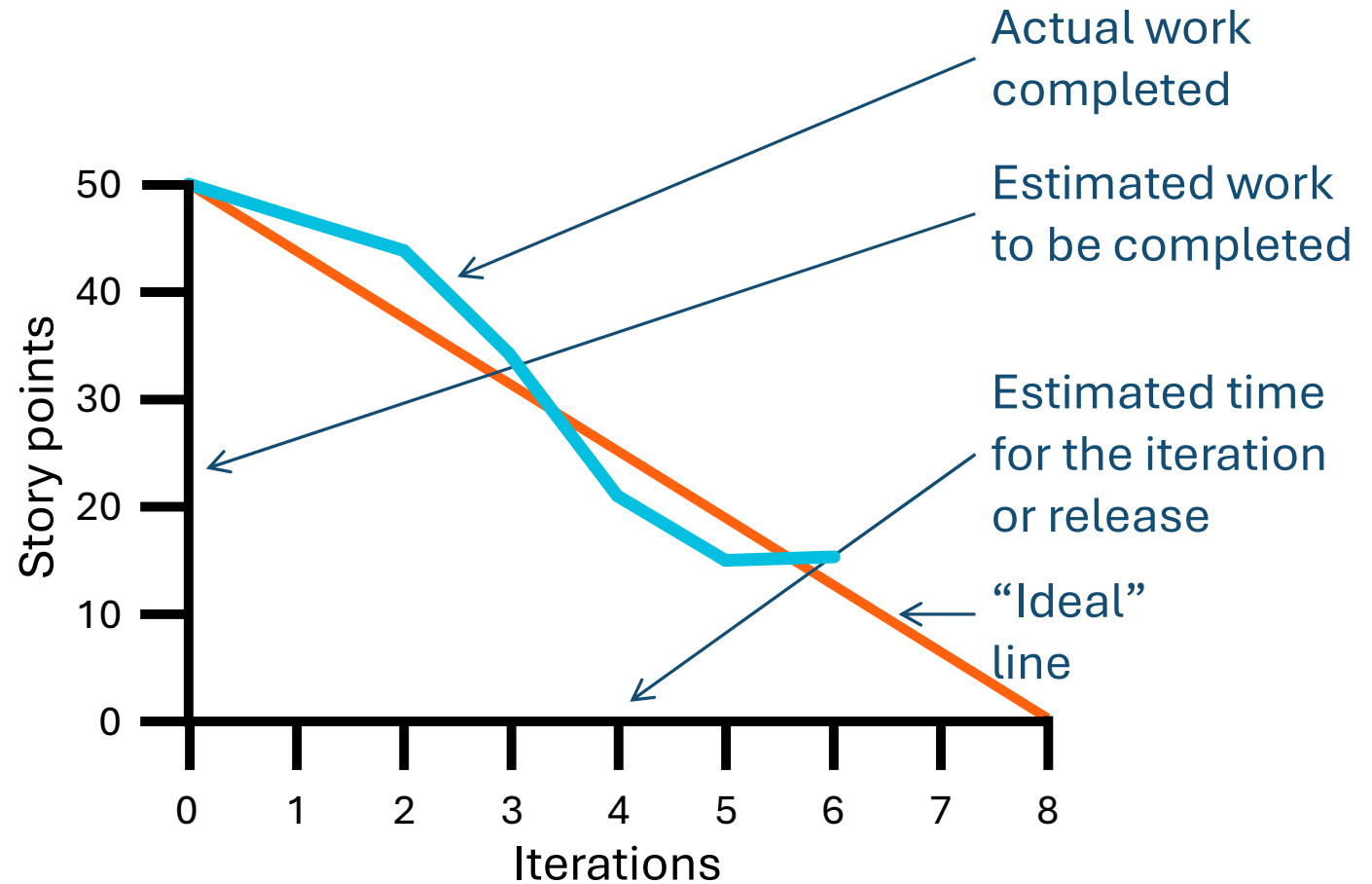
**Cumulative flow
diagrams**

Burndown chart

A **burndown chart** is a graphical representation of the work remaining versus the time left in a timebox.



Elements of a burndown chart



Advantages and disadvantages of burndown charts



Simple

Relevant to the goal

Future focused

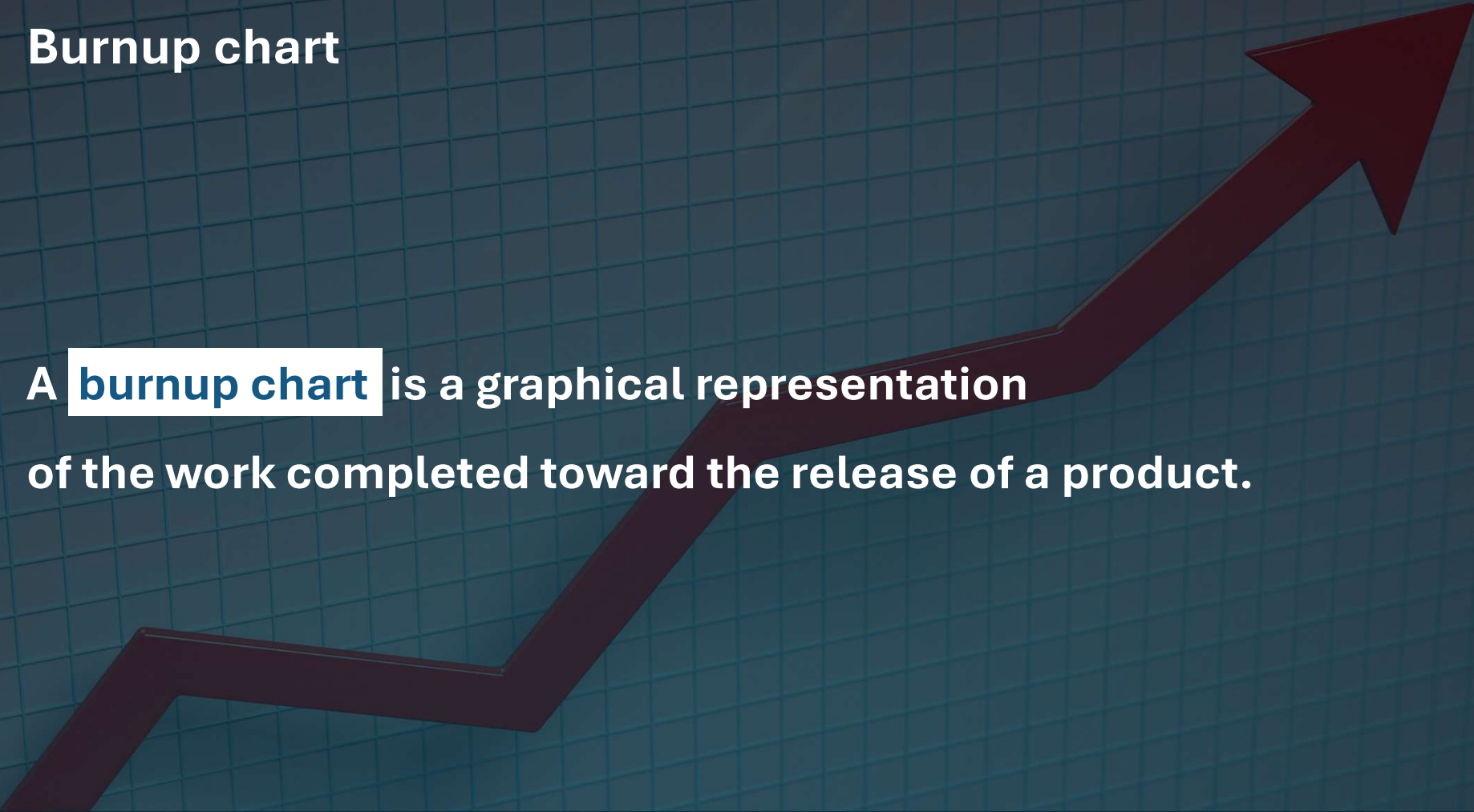


Often not automated

Doesn't reflect changes to
the scope

Burnup chart

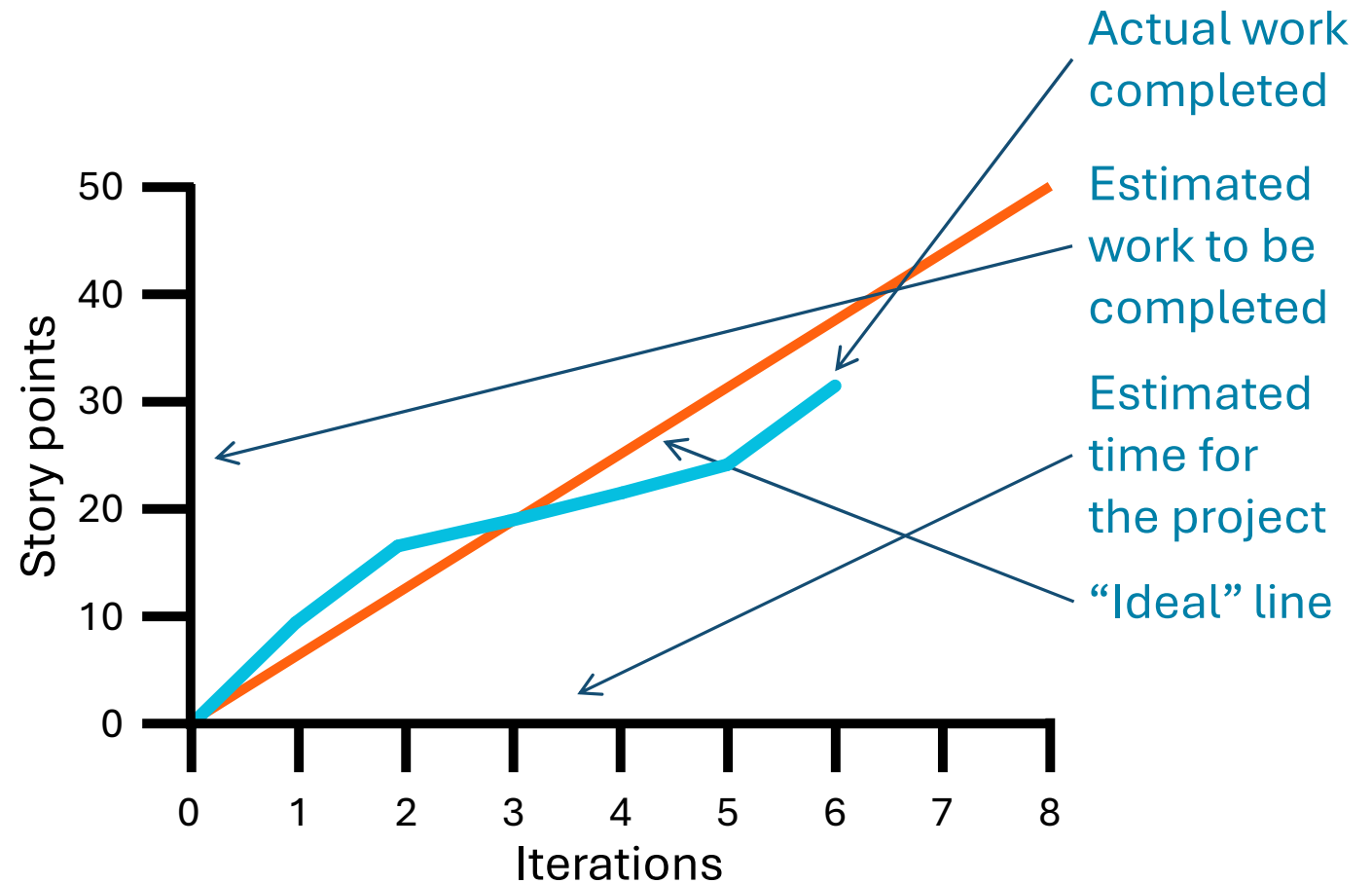
A **burnup chart** is a graphical representation of the work completed toward the release of a product.



Source: Project Management Institute. (2017). *Agile practice guide*. PMI.

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Elements of a burnup chart



Advantages and disadvantages of burnup charts



Simple and readable

Clear view of progress

Clearly see scope increases



May not be automated

Do not show tasks still in progress

Cumulative flow diagram

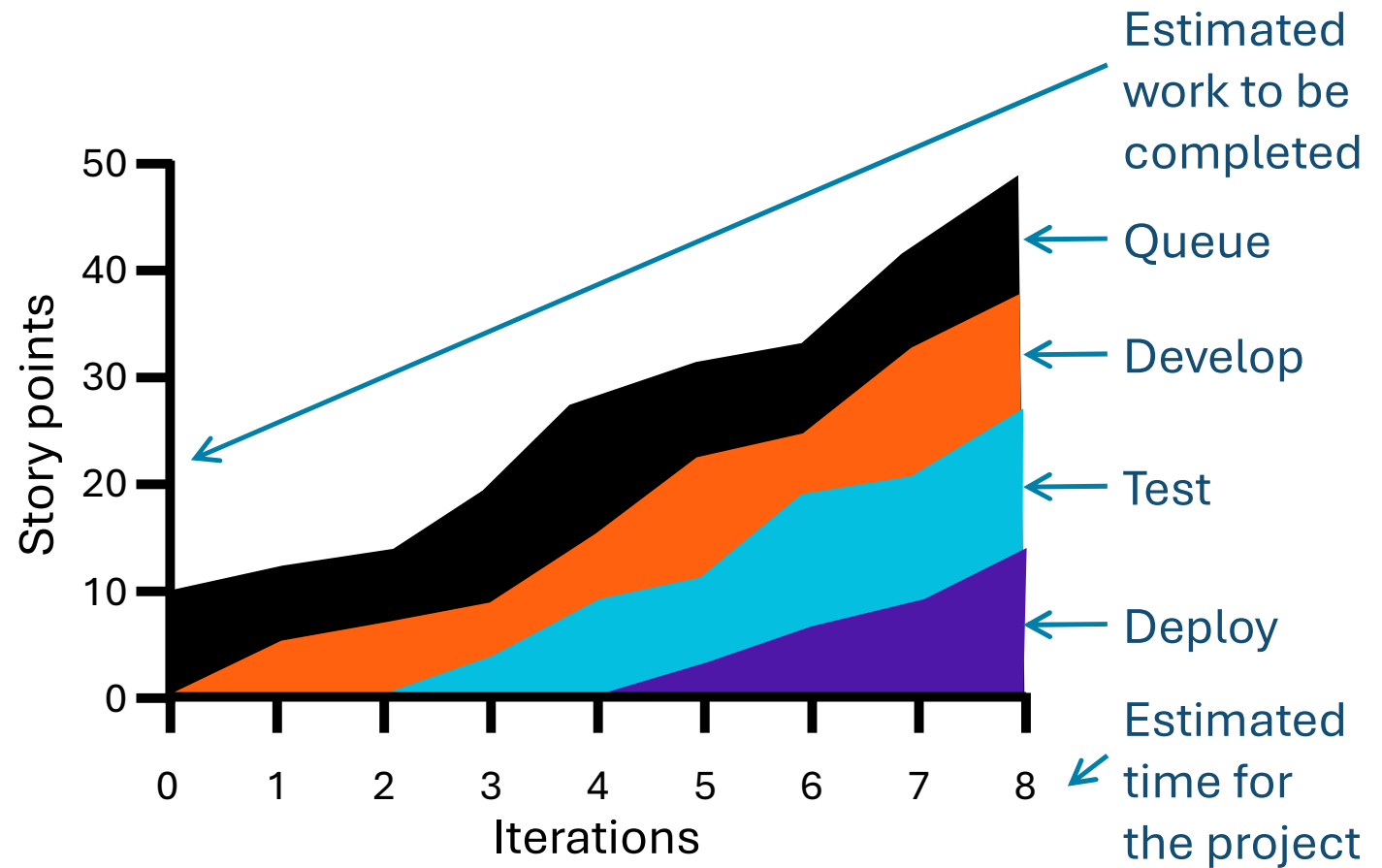
A **cumulative flow diagram** is a chart indicating the features completed over time, features in other states of development, and those in the backlog.



Source: Project Management Institute. (2017). *Agile practice guide*. PMI.

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Elements of a cumulative flow diagram



Advantages and disadvantages of cumulative flow diagrams



Shows rates of progress

Commonly generated by agile management tools

Can indicate potential bottlenecks



More complicated to generate

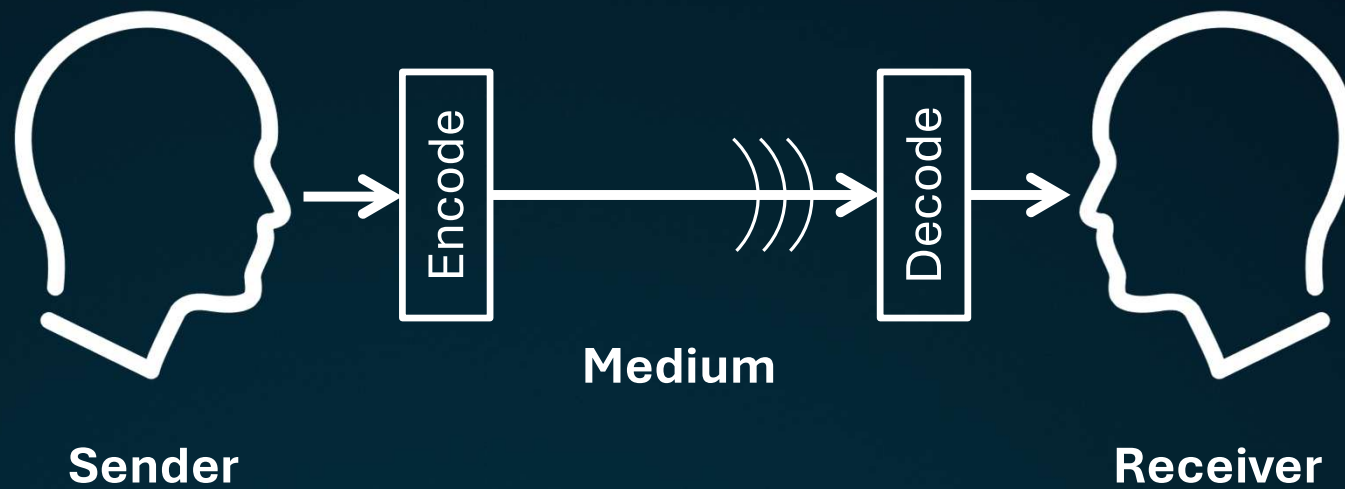
Requires some skill or explanation to interpret initially



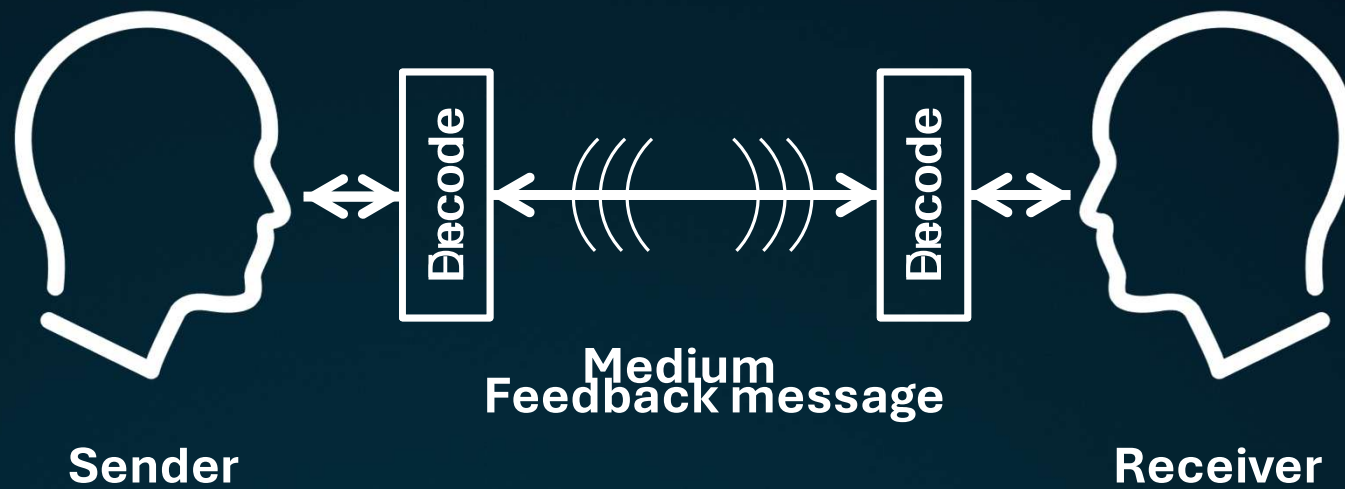
Radiate metrics across the relevant audience

Section 2 of 4

Basic communication model



Feedback



Applying the model to human communication



Framed in a way that is easily understood by the receiver



Appropriate medium



Minimize noise



Elicit feedback

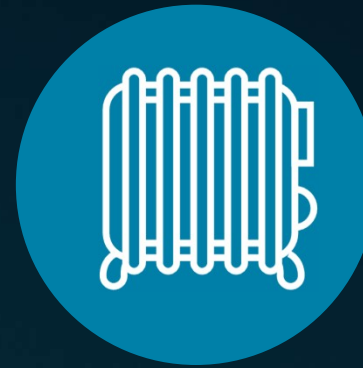
Communicating metrics to stakeholders



Push



Pull



**Information
radiators**



Demo

Push communications

Letters

Memos

Reports

Emails

Faxes

Voicemails

Press releases





Pull communications

People go to an information source and get the data they need

Information radiators

Displays information in a place where people have easy access to it



Examples of information radiators



Product vision



Sprint backlog



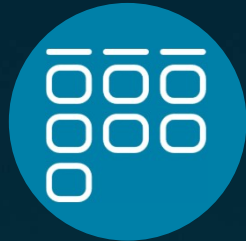
**Burnup and
burndown charts**



Product backlog



Impediments list



**Task/kanban
board**



Demos

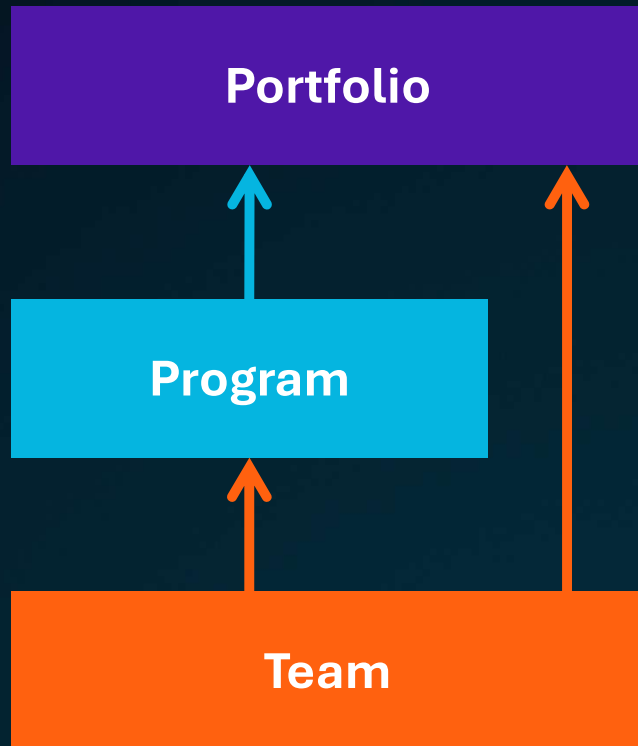
An excellent, regular
opportunity to show your
team's metrics

Rolling up metrics

The process of aggregating metrics from multiple sources—in summary form—to provide a broad view at a high level

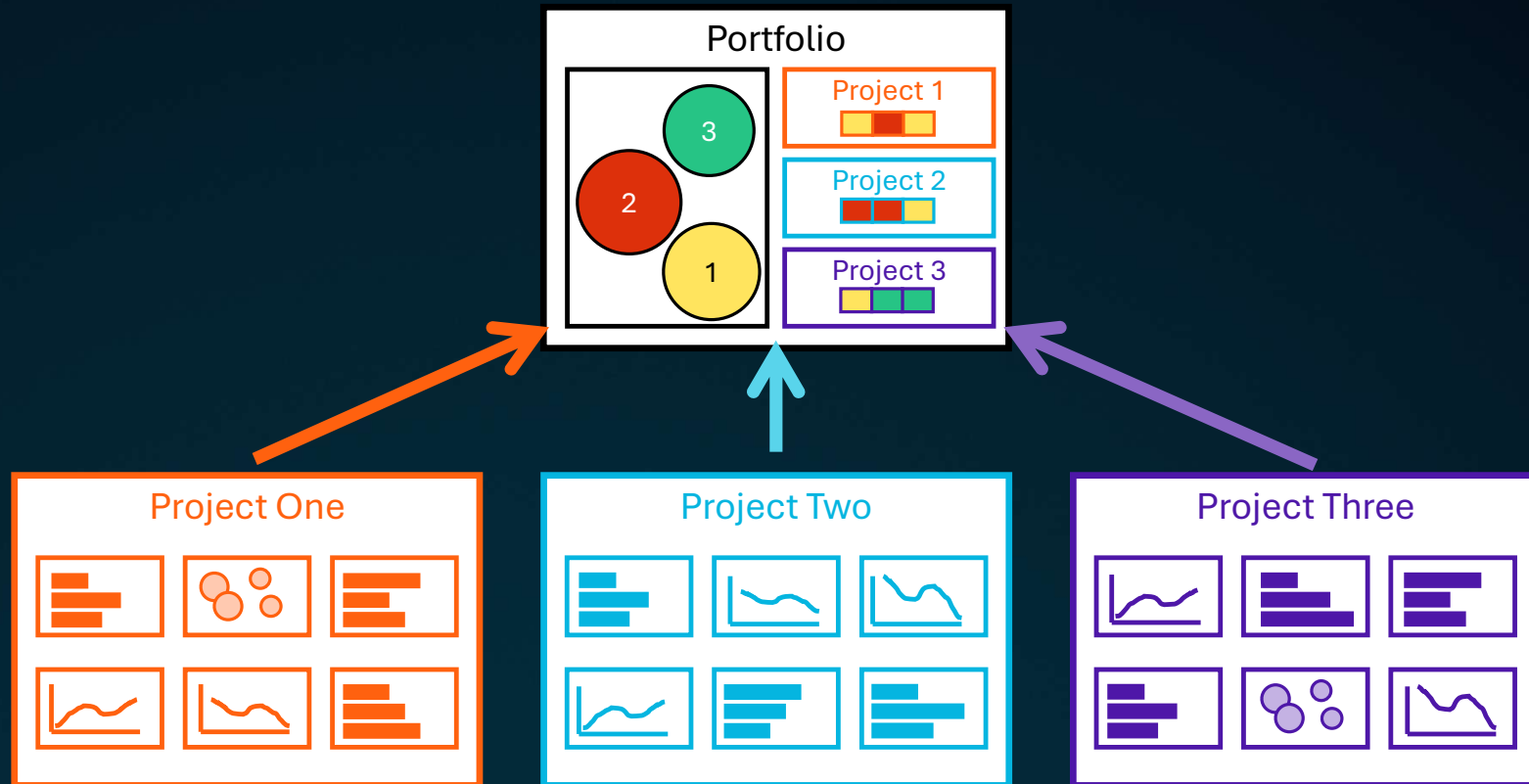


Metrics at different levels of the organization



- Measure against organizational goals
 - Overall indicator or category indicators
 - Organizational goals evolve slowly
-
- Measure against program goals
 - Category indicators per subteam
 - Program goals evolve periodically
-
- Measure against team goals
 - Specific metrics for each team
 - Metrics evolve as the situation evolves

How do we roll up metrics that aren't consistent among teams?





Review and analyze metrics

Section 3 of 4

Review sponsor and customer feedback





Review product owner feedback

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This material is being provided as part of a PMI course.

Review team feedback



Project metrics and the six project controls



Budget



Schedule



Scope



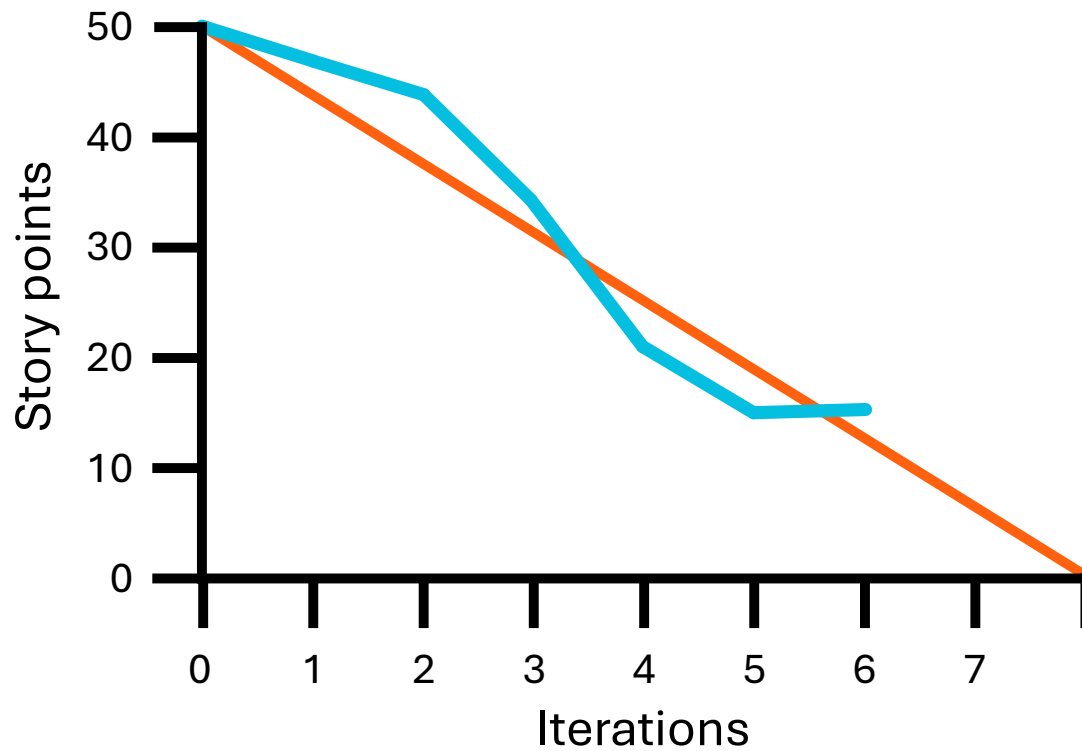
Risk



Quality

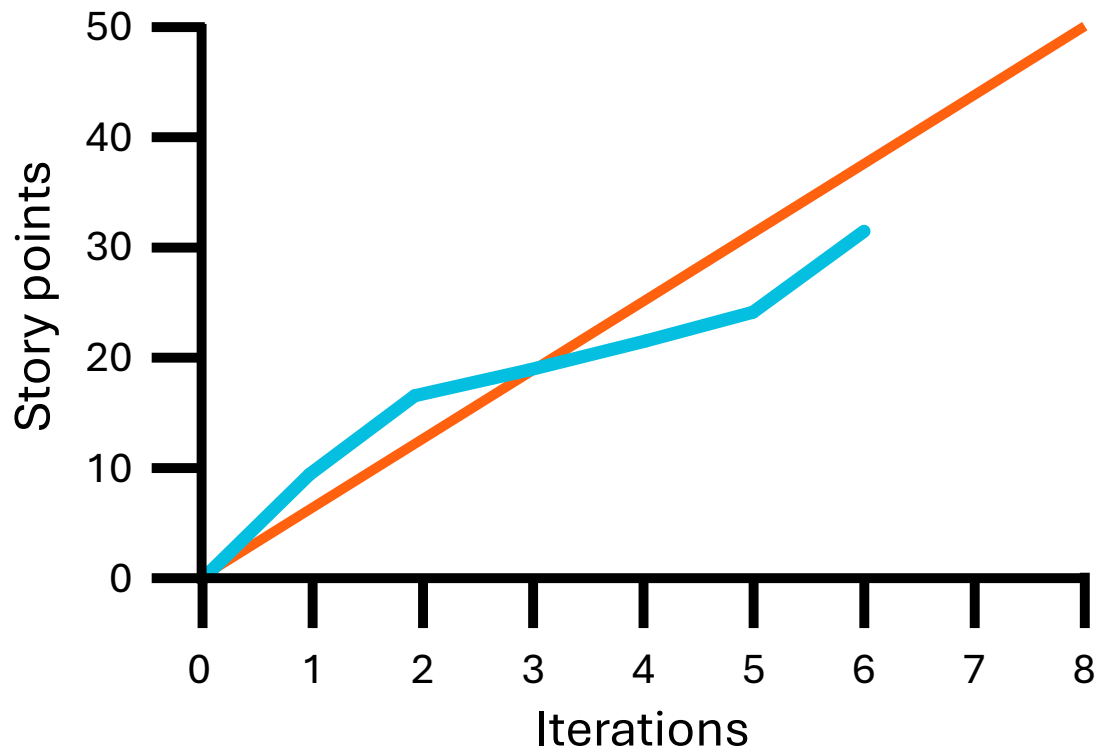


People



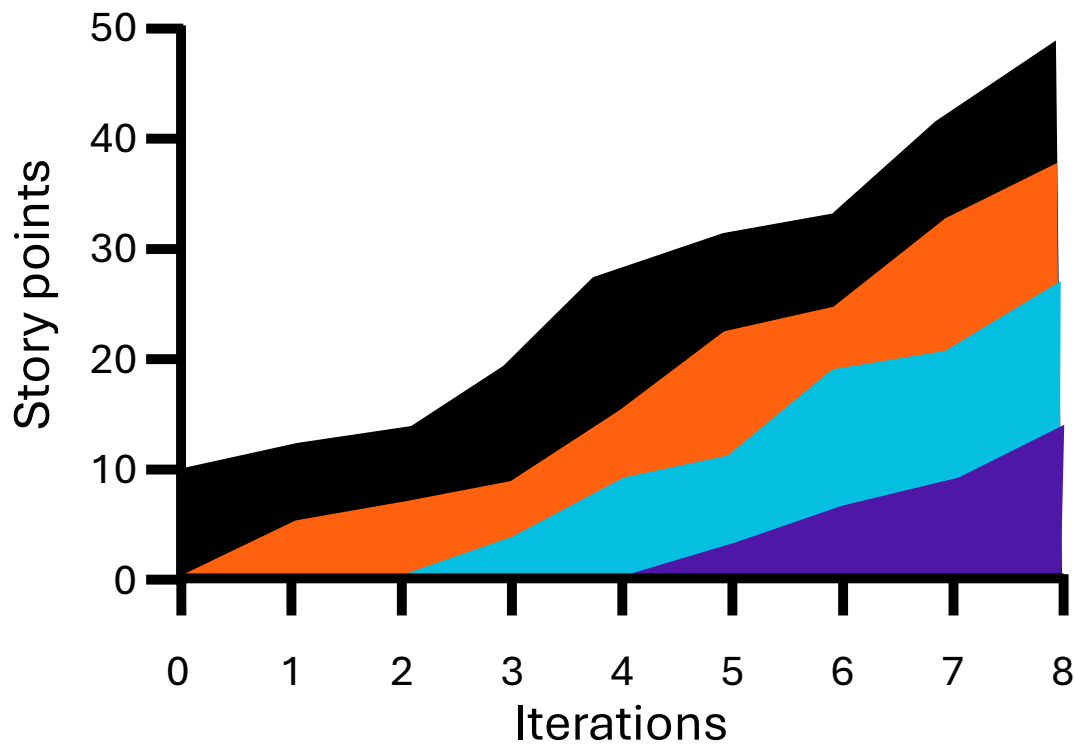
Burndown charts and the six project controls

- Budget
- Schedule
- Scope
- Risk
- Quality
- People



Burnup charts and the six project controls

- Budget
- Schedule
- Scope
- Risk
- Quality
- People

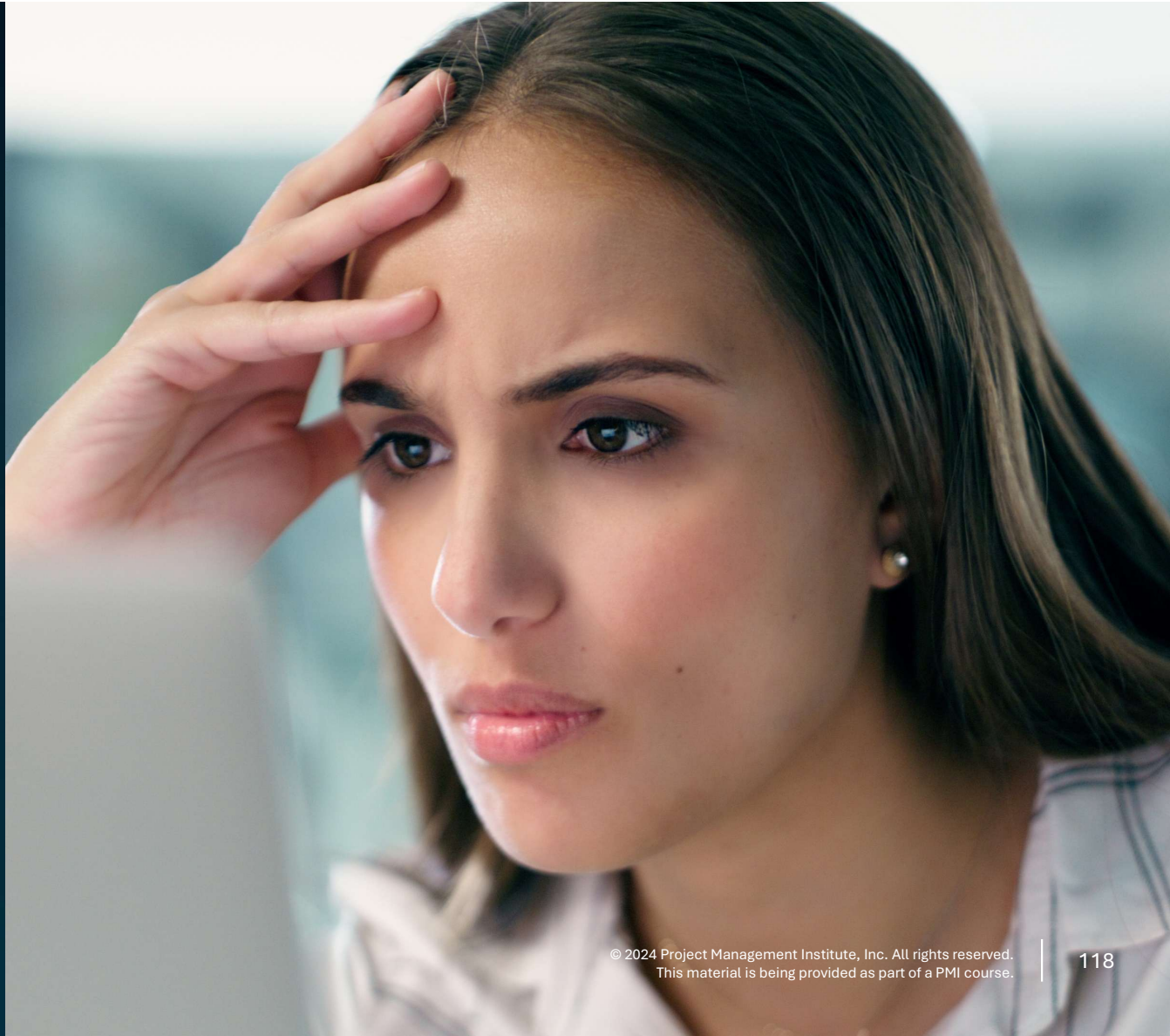


Cumulative flow diagrams and the six project controls

- Budget
- Schedule
- Scope
- Risk
- Quality
- People

How do you cover the other project controls?

You track other metrics



Additional metrics



**Budget versus
actual**



**Risk
burndown**



**Defect
density**



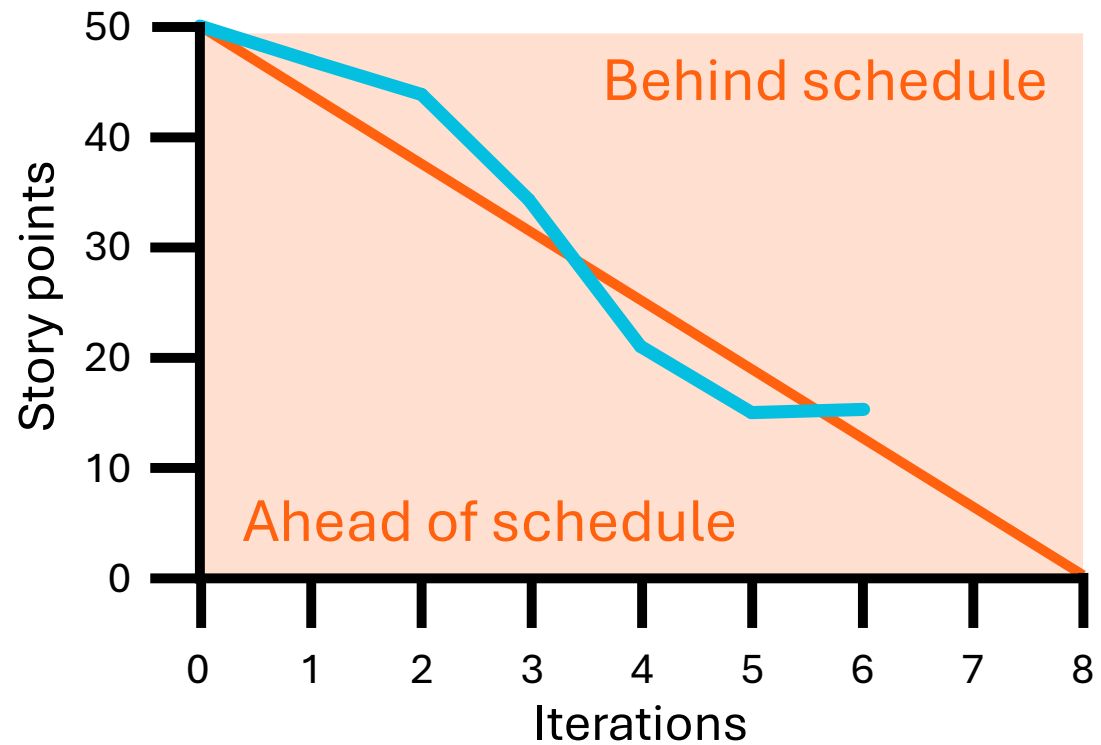
**Niko-niko
calendar**



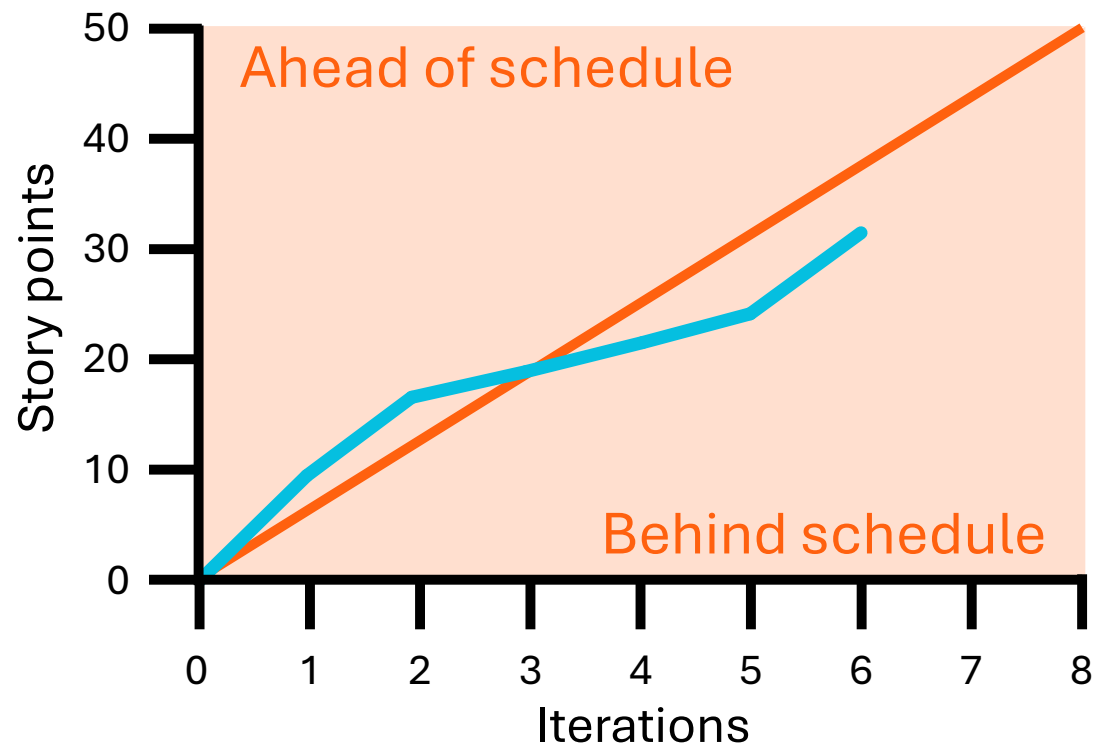
Analyzing metrics

How do you read and analyze the metrics your team is keeping?

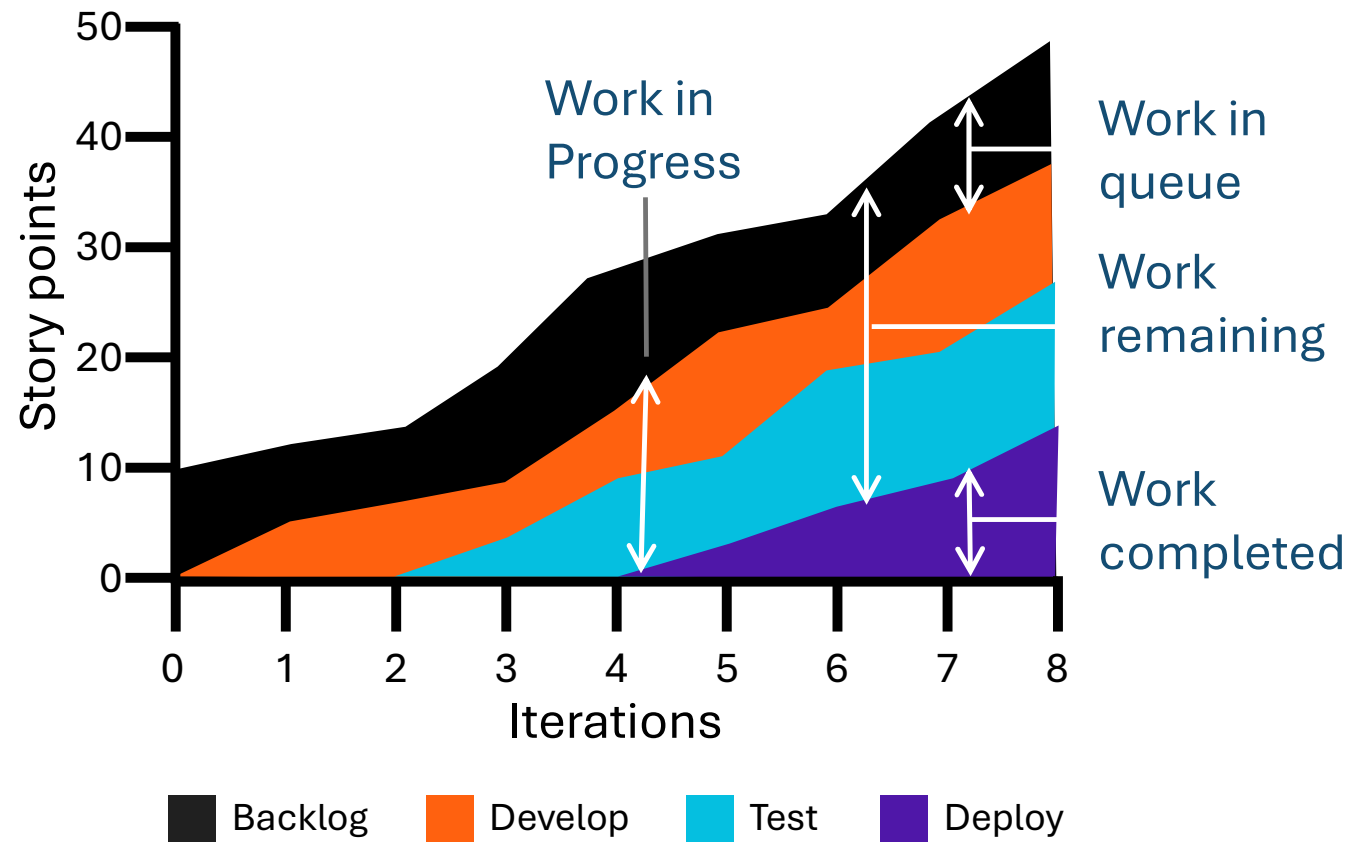
Reading a burndown chart



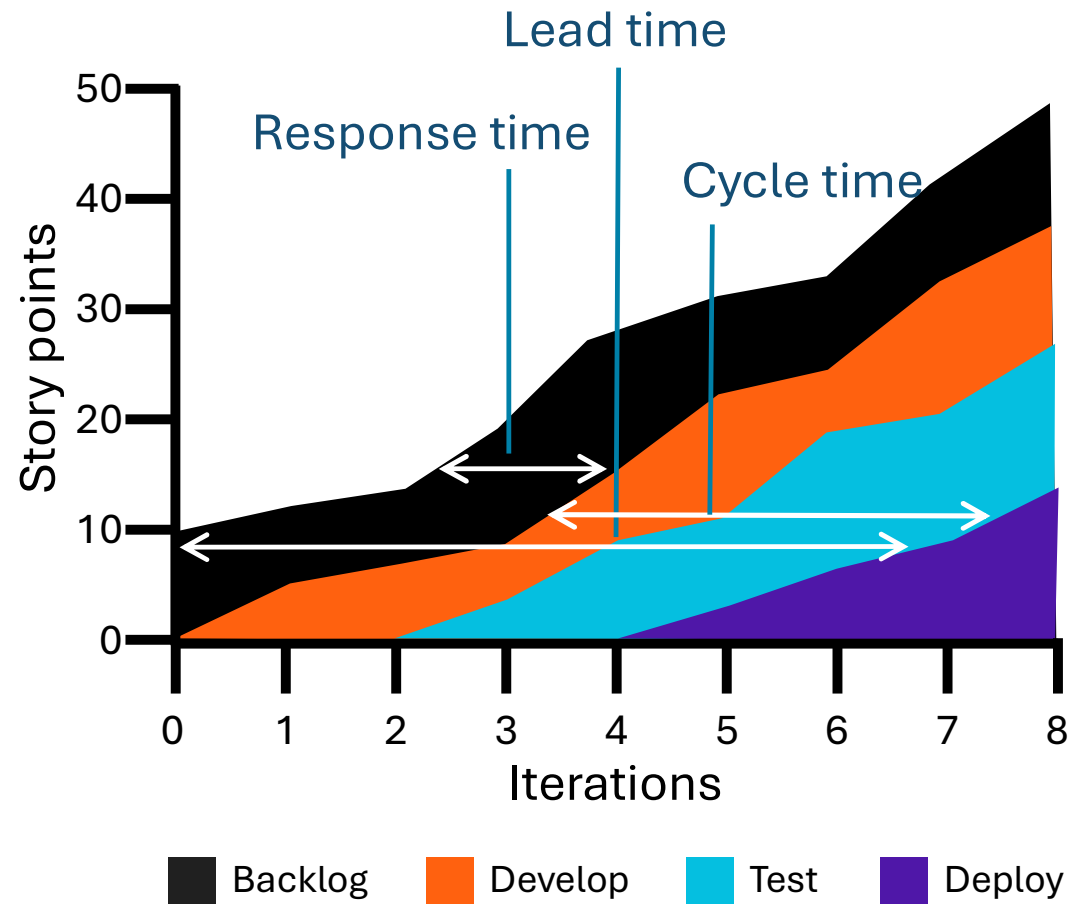
Reading a burnup chart



Reading a cumulative flow diagram



Reading a cumulative flow diagram



Knowledge check

What can you assume about the team's progress based on the burndown chart shown below?



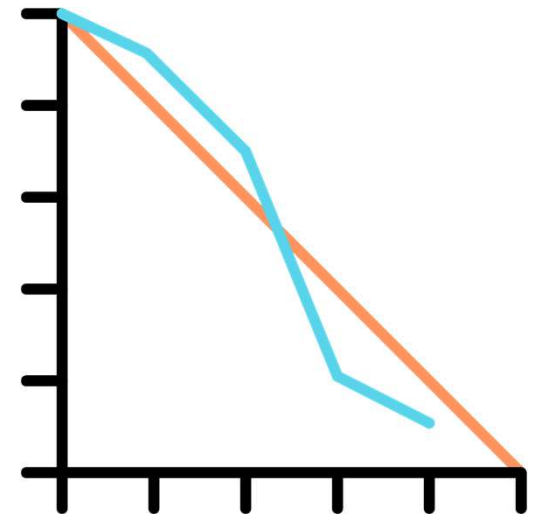
It is ahead of schedule



It is behind schedule



It is right on track



Since the team's work line is below the ideal line, we can assume that the team is **ahead of schedule**.

Knowledge check

What can you assume about the team's progress based on the burnup chart shown below?



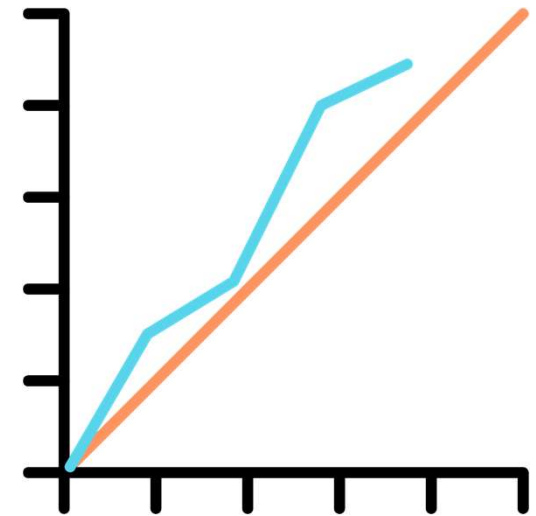
It is ahead of schedule



It is behind schedule



It is right on track



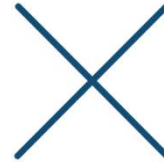
Since the team's work line is above the ideal line, we can assume that the team is **ahead of schedule**.

Knowledge check

True or false: Pull communications are those designed to pull stakeholders into a dialogue.



True



False

Knowledge check

True or false: One way to report on teams with different metrics is to use a stoplight status.



True



False

DAILY PMP BOOTCAMP SURVEY



LOOK FOR THE SURVEY LINK IN THE CHAT

- **Our goal** is to provide the best possible Bootcamp experience for a live streaming webinar, with hundreds of participants.
- For each Bootcamp session,
 - Let us know **what you liked** about the experience – your comments really matter.
 - Please include a thank you **to the mentor(s)** working off camera.
 - If you have **recommendations**, share those too!

We sincerely value your opinion!

Survey Scale

- This Scale: 0 not at all likely- 10 extremely likely



On a scale of 0-10, how likely are you to recommend this bootcamp to someone else?

This Scale: 0 not at all likely - 10 extremely likely

0	1	2	3	4	5	6	7	8	9	10
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐